GY457
Applied Urban and Real Estate Economics—*WT*

Course manager: Prof. Christian Hilber
London School of Economics

Lecturer’s introduction

- **Background**: Basel – Wharton / U. of Pennsylvania (Philly) – Fannie Mae (DC) – LSE (since 2003)
  - Professor of Economic Geography & Deputy Head of Department (Teaching) @LSE
  - Director of MSc REEF from 2003-2018 & Associate at CEP & Prof. of Real Estate Finance & Economics at U of Zurich (20%)
- **GY457** (Course Manager) (+GY210)
  - Role of government in real estate markets (Block I of WT)
  - Urban property markets and price dynamics (Block II of WT)
  - Housing issues and policies (Block III of WT)
- **Office Hours** during WT
  - Mondays 16:15—17:45, CKK.4.22 (sign up via LSE Student Hub)
  - Email: c.hilber@lse.ac.uk

Lecturer’s introduction – *Cont.*

- Care about my teaching and students – make use of office hours!
- Research interests – **Urban and real estate economics**
  - Functioning of land and property markets
  - Land use regulations: determinants & consequences
  - Property price & rent dynamics
  - Homeownership: determinants & consequences
  - Policy evaluation (mainly: housing & tax)
  - Role of housing markets for wealth inequality

Lecturer’s expectations

- **Be active during lectures & seminars** – why?
  - Importance of two-way communication (feedback)
  - Attention span curve
- Outside classroom
  - Discuss material/work **in groups**
  - Do not just memorize facts—**reflect & be critical**
  - Link theory to practice
- Why?
  - Research on effective learning suggests… importance of group discussions, collaboration & reflection
  - ‘Deep approach’ outperforms ‘surface approach’

Some organizational points:
Lectures

- **Syllabus** contains details on lecture schedule, seminars & readings (incl. what is compulsory)
- Lecture **handouts** available before lecture – **full lecturer slides** available after

Some organizational points: Seminars

- Seminars **differ in style** (debates, case studies, past exam questions etc.)
  - Led by myself (default) & Dr. Ignacio Aravena-Gonzalez (Weeks 5)
- Each seminar will have **presentations**
  - **Upload via Moodle** prior to the seminar
- Seminar preps usually involve **group work**
  - Responsibility of presenters to contact group for inputs/allocate tasks
  - If feasible: Arrange meeting to discuss (social aspect!)

Some organizational points: Exam preparations

- Will review exam format etc. during WT
  - **During seminar(s)**: Look at past exam paper & discuss exam format etc. + how to answer questions (Sem. 5 in W9)
  - **Mock exam** (voluntary): likely in Week 8 – details to follow
  - **Revision session** in Week 11: Exam format & exam preparation advice, Q&A & feedback on mock exam for those who take part
- **Marking criteria** – available on Moodle

Introduction –
Winter Term Lectures on **Real Estate Economics**

Introduction and Outlook

- **Urban economics**
  - The study of location choices of HH & firms within an urban area & institutional setting
- *Last term:* Focus on role of location choices of HH and firms (& equilibrium outcomes)
- *This term:* Focus on role of **institutional setting/government** (BLOCK I)
- Next: Focus on **real estate markets (**Real Estate Economics)
  - Housing (and commercial) markets (BLOCK II)
  - Housing issues, policies and policy evaluation (BLOCK III)

Contents—*Topics*

BLOCK I: THE ROLE OF THE GOVERNMENT

Topic 1: Interlinkages between local governments, public finances & housing markets (Week 1)

Topic 2: Land use regulation (Weeks 1-2/3)

- What is land use regulation? Why regulate?
- Determinants of restrictiveness
- Economic impact: benefits, costs & net welfare effects

Contents—*Topics (cont.)*

BLOCK II: URBAN PROPERTY MARKETS & PRICE DYNAMICS

Topic 3: Identifying housing supply and demand curves (W2/3-4/5)

- Identification problem
- Estimate supply and demand price elasticities
- Two-Stage Least Squares (2SLS)

Topic 4-6: Real estate cycles (W4/5,W7-W9)

- Stylized facts & theory (Topic 4)
- Exogenously driven cycles (Topic 5)
- Endogenously driven cycles (theory & evidence) (Topic 6)
- Price vs. rent dynamics
- Residential vs. commercial cycles

Contents—*Topics (cont.)*

BLOCK III: HOUSING ISSUES & POLICIES

Topic 7: Owning vs. renting: Externalities, determinants & policies (W9/10)

- Stylized facts
- (Why) Should we care about homeownership attainment?
- Should governments subsidize homeownership?
- Determinants of tenure choice/homeownership
- In seminar only: Are homeownership subsidies (like MID) effective?

Contents—*Topics (cont.)*

BLOCK III: HOUSING ISSUES & POLICIES

Topic 8: Housing issues in China (W10-11, with Xiaolun Yu)

- Stylized facts
- China's affordability crisis: possible causes
- Policies and policy evaluation

Revision Session (Week 11: Wednesday 2-4:30pm)

Block I—**Topic 1:**

Local Governments, Public Finances and Housing Markets

How do governments influence land and real estate markets? (Discuss)

- Directly
  - Housing policies (e.g., HtB in UK; cooling measures in China)
  - Land use restrictions including zoning rules
- Indirectly
  - Enacting policies & regulatory environments that affect location choices
  - Providing (supra-)national public goods/services & taxes
  - Local public goods (LPGs)/services and taxes

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Two observations

1. Within countries, local house prices are mainly determined by
   - Agglomeration economies *(London vs. Reading)*
   - Local amenities *(microclimate, historic cityscape, lakes, rivers, nice views etc.)*
   - Quality of local public goods (LPG) *(schools!)*
   - Tax price for LPGs
2. In some countries LPG are provided by central government, in others locally

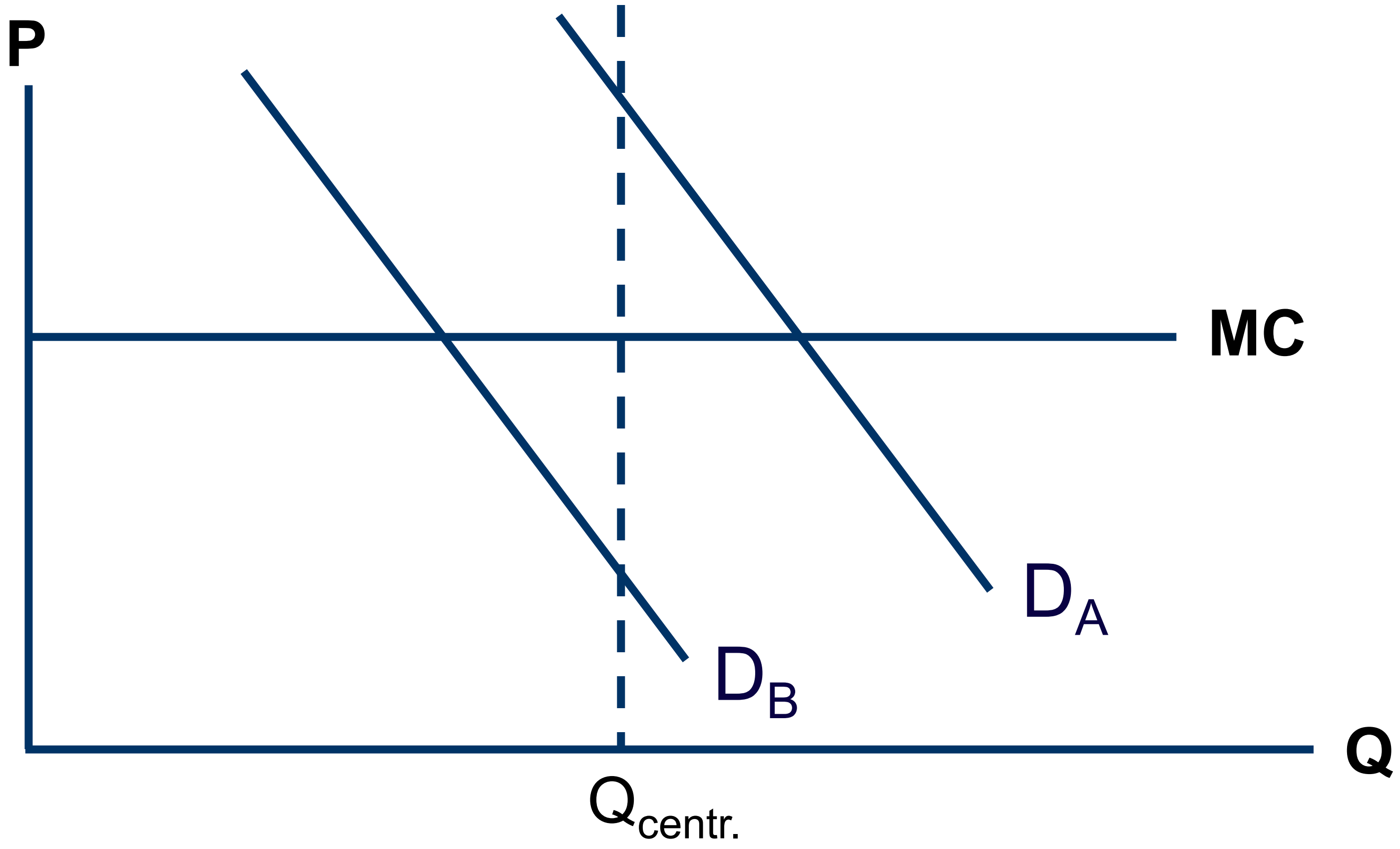
⇨ **Which is better?**

Centralisation versus decentralisation

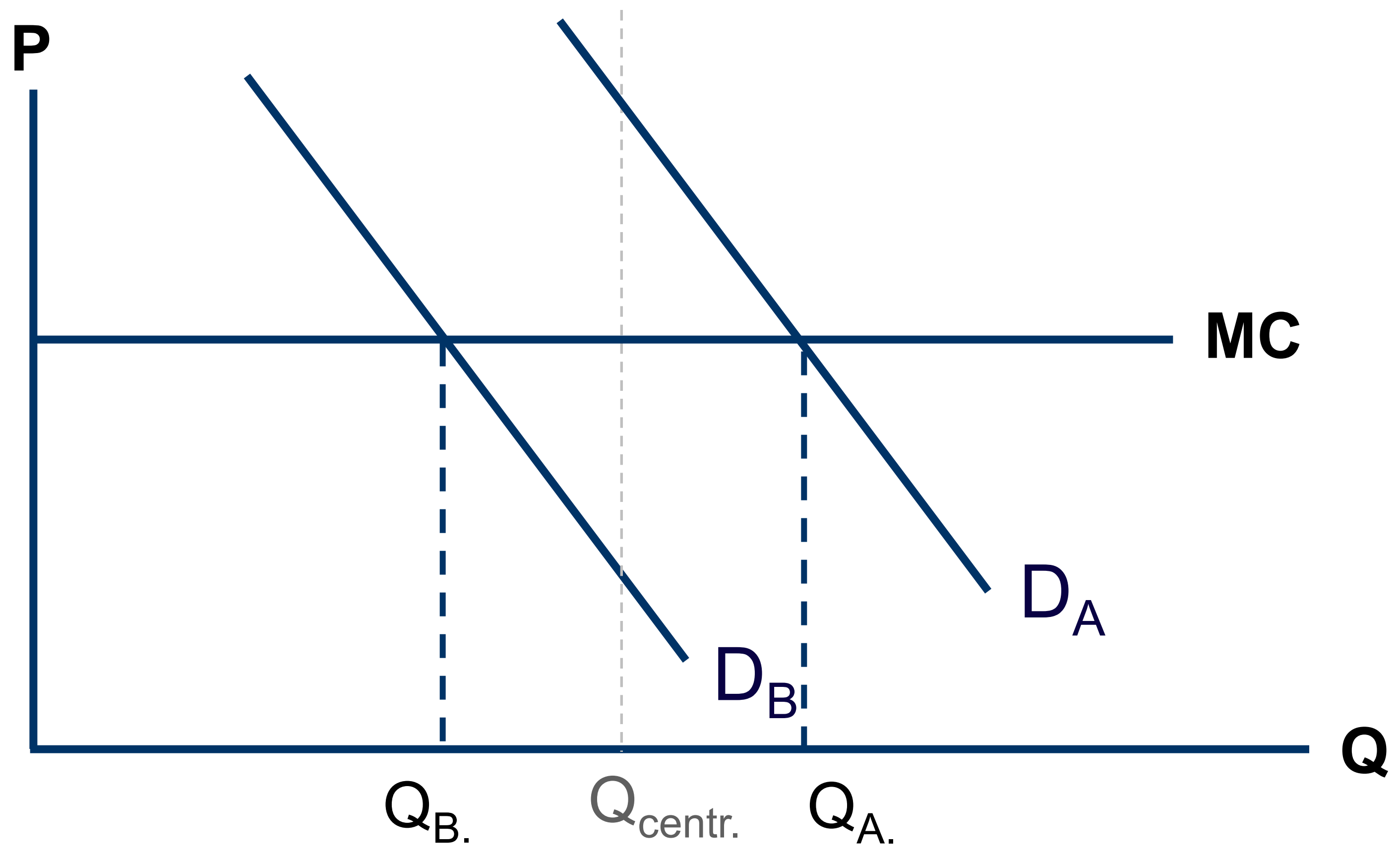
- Consider
  - 1 Local public good (non-rival & non-excludable)
  - 1 Central government
    - Social planner that faces uniformity constraint
  - 2 Local jurisdictions
    - Community A has strong demand for public good
    - Community B has weak demand for public good

⇨ **Should the public good be provided by central government or at local level?**

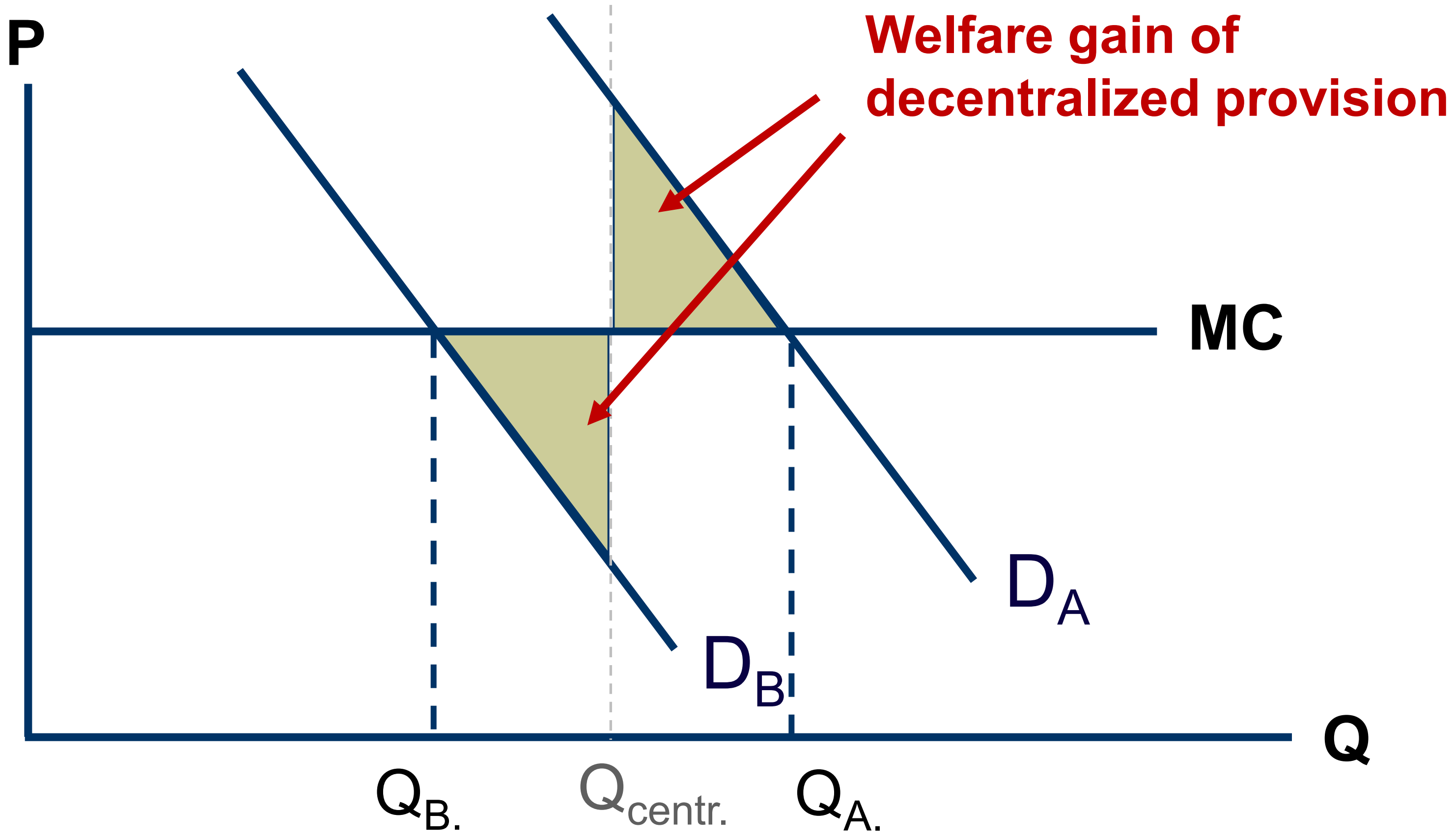
Centralized provision



Decentralized provision



Which is better?



Centralisation versus decentralisation *(Cont.)*

- Proposition
  - Decentralized provision allows differentiated provision of LPG and thereby avoids welfare losses
- Critique? *(Discuss)*
  - **Economies of scale** in production?
  - How do local governments know about the **preferences** of their residents?
  - How to reveal preferences? Doesn't everybody have an incentive to behave as **free-rider**?
  - Income inequality & sorting – **fairness** considerations

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Is there a market-mechanism that leads to optimal provision of public goods?

- Samuelson (1954)
  - No viable mechanism to reveal individual preferences for public goods
  - Private agents try to behave as free-riders
  - No private agent wants to provide public goods

⇨ Samuelson's answer is **NO!**

Really true?

- Tiebout (1956)
  - Individuals sort themselves across local jurisdictions according to their preferences for local public goods
  - Residents thereby reveal their preferences for local public goods
  - “Spatial mobility provides local public counterpart to the private market’s shopping trip”

Tiebout's (1956) model

- Prototypical Tiebout World
  - Assumes perfect knowledge of residents, costless relocation, a large number of communities & an endogenous community size
  - Residents perfectly sort into communities with their exact preferences
  - Example: Elderly households move to communities that focus exclusively on services for elderly

Tiebout's (1956) hypothesis

- **Consumer mobility** and **inter-jurisdictional competition** lead to ***efficient*** provision of local public services
- Reasoning?
  - Residents reveal true preferences for local ('distance sensitive') public goods through '**voting with the feet**'
  - Profit-maximizing private entrepreneurs/local administrators use this information to provide an optimal level of public services

Critique? *(Discuss)*

- Model based on unrealistic/restrictive assumptions
  - Costless mobility
  - Perfect knowledge
  - Large number of competing local jurisdictions
  - Endogenous community size
  - No external economies or diseconomies
  - No multidimensional preferences
  - No land or labour market
  - No commercial RE
- Does Tiebout-hypothesis still hold?
- Can we test empirically?

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Tiebout sorting and house price capitalization

- Oates (1969)
  - If consumers shop among local communities, then fiscal differentials will be capitalized into house values (**Capitalization Hypothesis**)
  - Interprets capitalization as evidence for Tiebout-hypothesis

Can capitalization persist in equilibrium?

- Edel & Sclar (1974) and Hamilton (1976)
  - In world with completely elastic supply of local communities, public sector variables will be uncorrelated with house prices in full Tiebout equilibrium
  - Capitalization is disequilibrium phenomenon that will disappear in the long-run
- Epple & Zelenitz (1981)
  - Capitalization can exist in equilibrium if community boundaries are fixed exogenously

Main conclusion of theoretical literature

- Existence of capitalization is consistent with view that consumers ‘shop’ among local communities
- BUT…Existence of capitalization **does not test the Tiebout-hypothesis**
- Capitalization effects arguably still important – why? *(Discuss)*
  - Important **distributional effects** (homeowners vs. renters)
  - Provides **incentive effects**, e.g., to invest in better local schools

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House price capitalization —Empirical evidence

- Oates (1969) finds empirical evidence for capitalization
  - Positive and large effect of school expenditures on house prices
  - Increase in PV* of property tax by $1000 reduces HP by $664
  - Uses OLS and TSLS
- Results confirmed by many later (much more refined) studies

* Say tax increase is $50, and the discount rate is 5%, then PV = $1000 = $50/0.05 (of course there is also uncertainty about 'true' discount rate & whether tax increase is permanent...)

More recent (rigorous) studies on capitalization of school quality

Study	*Impact on Prices*
Gibbons *et al.* (2013 JUE)	3% per 1 std. dev.
Clapp, Nanda and Ross (2008 JUE)	1.3-1.4% per 1 std. dev.
Fack and Grenet (2010 JPUB)	2% per 1 std. dev.
Davidoff and Leigh (2008 EcRe)	3.5% per 1 std. dev.
Brasington and Haurin (2006 JRS)	7.1% per 1 std. dev.
Kane, Staiger and Riegg (2006 AmLawEcRe)	10% per 1 std. dev.
Gibbons and Machin (2003 JUE; 2006 EJ)	3.8-10% per 1 std. dev.
Cheshire and Sheppard (2004, EJ)	4-10% per 1 std. dev.
Rosenthal (2003 OxBuEcSt)	5% elasticity
Black (1999, QJE)	2.5% per 1 std. dev.

Most recent (rigorous) study on capitalization of local taxes

- Basten *et al.* (2017, EJ) explore impact of **local income tax differentials** (in Switzerland) on housing rents
- Employ **BD-approach** that corrects for unobservable location characteristics (incl. holding school district constant)

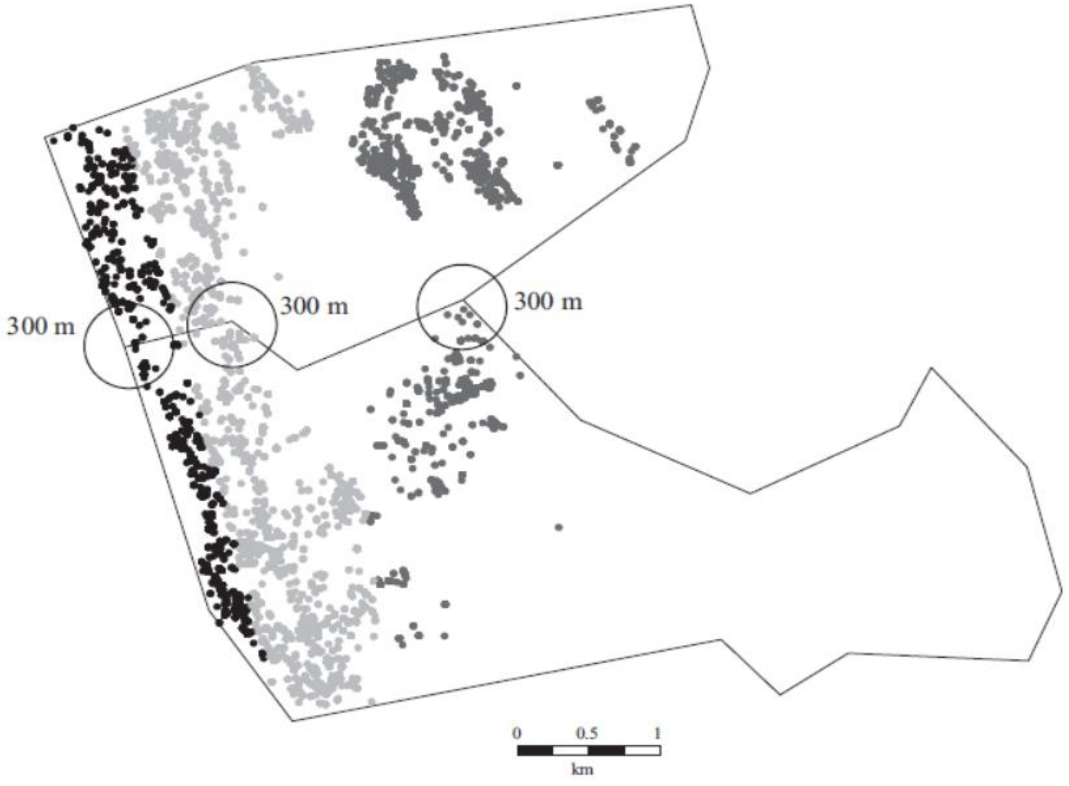


Fig. 5. *Residences and Border Points – Example: Zollikon/Küsnacht*

Tax capitalization—*cont.*

- **Income tax elasticity of rents ~0.3** (10% ↑ in taxes → 3% ↓ rents) corresponds to ~50% capitalization*
  - Significantly lower compared to estimates from conventional approach (hedonic regression)
  - Suggests important role of unobservable location and neighborhood characteristics

* Say income tax per month & HH = CHF1200, rent per month = CHF 2000,
then 10% increase in tax = +CHF120 ⇨ 3%↓ in rents = -CHF 60 ⇨ -CHF60/CHF120 = 0.5 (50%)

Capitalization of fiscal aid

- Hilber *et al.* (2011, RSUE) explore impact of **central government grants** in UK on house prices
- Employ **Instrumental Variable (IV)-strategy**
  - Idea: Employ exogenous source of variation in grants that is strongly correlated with grants but not directly related to house prices
  - Instrument: Exploits fact that incumbent central government targets extra grants to ‘swing districts’ (LAs) conditional on victory by own party
- Estimates suggest **substantial to full capitalization** of grants in house prices
  - Capitalization effects are greater in locations with physical barriers to construction

Policy implications? *(Discuss)*

- Capitalization of fiscal aid **benefits typically better off property owners** (homeowners & absentee landlords) in deprived areas
  - More generally: Capitalization effects–habitually ignored by policy makers—can have important **adverse consequences** for wide range of policies (e.g., fiscal aid, MID)

- But: Capitalization may also have **desirable incentive effects**
  - May induce investment into **durable** local public goods because benefits are capitalized into property values today (e.g., investment in new high-quality school)

Preparations for *Block I*—Seminar 1 (Week 3 WT)

- **Case study/debate**
  - Should countries *in transformation* be organized in a centralized or decentralized fashion?
  - What role do housing market conditions play?
  - Debate with two lobby-groups and some 'judges' (policy makers)

⇨ **Preparation: See separate handouts for sub-groups A, B and C**

⇨ **Sub-group allocation: See separate handout**

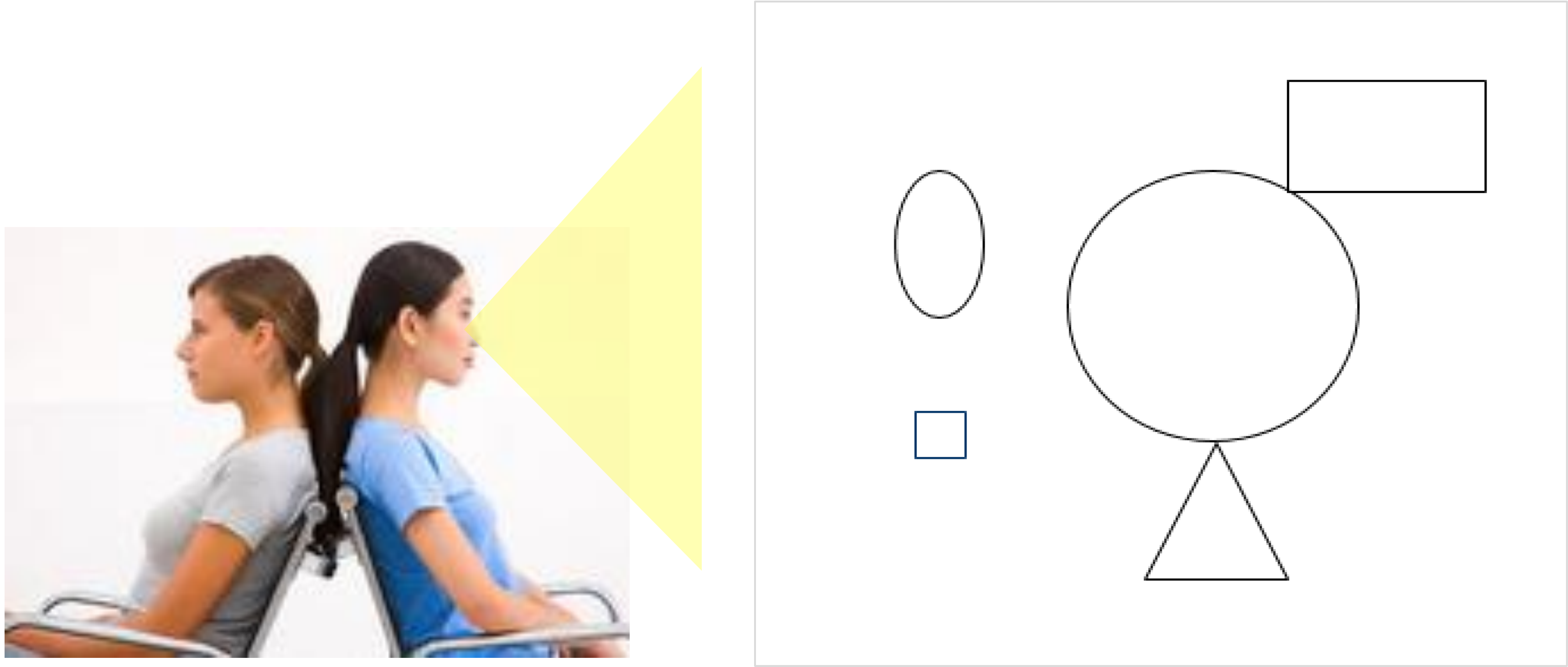
Organisational stuff

- **Presenters**
  - Default is always 2 students jointly
  - Number of presentations/presenters varies from week to week
    - Seminar 1 has 3 sub-groups in AM & PM so 2 x 3 x 2 = 12 presenters in total
- **Reminder:** Upload presentation **via Moodle prior to the seminar**
  - Responsibility of selected presenters in each week

Reflection

- Write down the 3 most important things that you have learnt today…

The experiment…



Appendix: Definition of **Real Estate Economics**

Real estate economics applies the tools of economic reasoning to the analysis of real estate markets. It explores how supply and demand interact to determine rents and prices within an institutional setting.